VENKATARAMAN SWAMYNATHAN

Lead R&D Engineer and **IIM Ahmedabad (AGMP)** candidate with over 12 years of global experience in the Telecom sector. Proven track record of spearheading complex system validation and R&D strategies for Tier-1 US carriers, including T-Mobile and AT&T. Currently bridging the gap between technical mastery and strategic general management by mastering scalable operational, marketing, and sales frameworks. Seeking to leverage deep-domain expertise in Software QA and high-stakes Project Management to drive organizational transformation and cross-functional excellence.

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CORE COMPETENCIES

- **Strategic Transformation & Change Management**
- **Project Planning & Life Cycle Management**
- **Stakeholder & Client Engagement**
- **Risk Assessment & Mitigation**
- **Team Building & Cross-Geo Collaboration**
- **Employee Growth & Accountability**
- **Telecom Protocol Testing (Diameter/SIP/Web-RTC)**
- **Revenue Assurance & Billing Systems**
- **Network Resiliency & Failover Testing**
- **Automation Strategy & CI/CD Integration**

PROFILE SUMMARY

- **Currently pursuing Accelerated General Management Program from Indian Institute of Management, Ahmedabad** (Online mode).
- **Over 10 years of extensive experience in QA & System Engineering** within the Telecom sector, specializing in VoIP, VoLTE, SIP Protocol, Diameter Protocol, IMS, and EPC
- **Driving end-to-end QA and performance initiatives at Telaverge Communications, ensuring products meet rigorous quality standards and are delivered on schedule.**
- **Spearheaded the testing initiatives for TMO WebClient (DIGITS 2.0) at T-Mobile**, guaranteeing flawless customer interactions during voice calls.
- **Demonstrate a deep comprehension of Software Quality Assurance methodologies**, SDLC and the Defect Life Cycle, ensuring high-quality deliverables throughout the software development process while following **Waterfall & Agile methodologies**.
- Rich experience in **rendering technical leadership & guidance to the team members, planning, scheduling tasks**, holding discussions with different functions / management.
- **Technical capabilities in managing complete STLC** entailing scripting, analyzing & implementing test scripts, test cases, test plans, test processes, defect reporting, bug tracking & report generation of different applications / products.
- **Proficient in Test planning, Test Strategy, Requirement Gathering & Test-case execution**, contributing to robust product quality with expertise in **Troubleshooting, Root Cause Analysis, and resolving defects**, enhancing product reliability.
- **Well-versed with modem/device certification processes**, ensuring compliance with industry standards and **knowledge of Session Analyzer – IRIS, QXDM/QCAT & JDSU**.
- Experience in **implementing company's QA** strategies with technical skills, both functional & non-functional, manual & automation, ideally in a continuous delivery environment.
- Excel in **development, design & validation of Test Cases, leveraging Use Case** & Functional Business Requirements to ensure comprehensive test coverage.
- Showcased excellence in **end-to-end Project Management functions** using Agile (Scrum) Methodology including Initiation, SOW, Program Scoping, Planning (Estimation, Scheduling & Milestones), KPI Monitoring, Risk management & Delivery.
- **People Leader, who has successfully led & motivated teams** in cross-cultural environment towards growth and success in the organization.

WORK EXPERIENCE

Oct'24 – Present: Telaverge Communication | Bengaluru, India

Lead Engineer – Research and Development

Architecting carrier-grade Diameter Stack validation and R&D strategies to drive industry-wide transformation. Expertly automating complex network scenarios to scale product reliability and accelerate CI/CD-integrated delivery for global telecom infrastructure.

- **Strategic Quality Leadership:** Direct the end-to-end validation strategy for high-performance Diameter Stack implementations across multiple telecom interfaces, ensuring 100% product reliability for global tier-1 carriers.
- **Technical Excellence:** Spearhead complex testing and validation using Seagull traffic generators and the Regal automation framework within Linux environments to deliver mission-critical firmware upgrades.
- **Process Transformation:** Drive CI/CD-integrated quality improvements by identifying high-impact automation candidates, significantly reducing manual regression cycles and enhancing time-to-market.
- **Lifecycle Management:** Architect comprehensive test suites for new features and enhancements, managing the full pre-release validation cycle from functional testing to large-scale performance benchmarking.
- **Continuous Improvement:** Champion QA process optimization through rigorous defect analysis and expanded test coverage, aligning engineering output with evolving market requirements and customer expectations.
- **Strategic Risk Mitigation (New):** Manage the deployment roadmap for firmware upgrades by conducting rigorous impact analysis, ensuring zero-downtime transitions and maintaining high system integrity for end-users.

- Large-Scale Subscriber Migration
- Device & OEM Certification
- Quality Assurance Strategy & Process Design
- Agile, Waterfall & Hybrid Methodologies
- Test Effort Estimation & Resource Optimization
- Root Cause Analysis (RCA) & Defect Prioritization
- Customer Satisfaction & QoS Benchmarking
- Cross-Functional Business Literacy

TECHNICAL SKILLS

Diameter | Linux |VoLTE | LTE | IMS | EPC | API Testing | REST API Testing | Session Analyzer – IRIS | QXDM /QCAT | JDSU | Root Cause Analysis | Debugging |

EDUCATION

- **(2026)**: Pursuing Accelerated General Management Program from Indian Institute of Management, Ahmedabad
- **2011:** M.Sc. in Electrical Engineering with Specialization in Wireless Communication from LC Smith College of Engineering and Computer Science, Syracuse, NY
- **2009:** BE in Electronics & Telecommunication from Atharva College of Engineering, Mumbai, India

Apr'20 – Aug'23: Orchestra Technology Pvt. Ltd.| Chennai, India | Client: T-Mobile

Senior Software QA Engineer

Spearheaded the end-to-end quality lifecycle for T-Mobile's DIGITS 2.0, enabling high-availability voice services across web and desktop platforms. Managed critical 911 regulatory compliance and presented system-resilience roadmaps to executive leadership to minimize downtime.

- **Product Leadership:** Led the end-to-end quality assurance for T-Mobile's DIGITS 2.0 (Web & Desktop), managing the WRG node integration to enable seamless browser-based voice and messaging services.
- **Regulatory & Compliance Oversight:** Orchestrated the implementation and validation of critical Emergency Calling (911) functionalities, ensuring strict adherence to federal telecom regulations and public safety standards.
- **Automation Strategy & Management:** Spearheaded the development of a customer-centric Automation Test Suite, replicating complex user journeys to preemptively identify failures and ensure an error-free production environment.
- **System Resilience:** Documented and presented strategic system enhancements to executive management, driving the adoption of proactive measures that significantly reduced system downtime and improved platform stability.
- **End-to-End Quality Lifecycle:** Managed the full testing lifecycle—including regression, sanity, and feature-specific validation—for high-impact functions such as Voicemail, Messaging, and Do Not Disturb (DND).
- **Requirements Engineering:** Translated complex business requirements into robust test architectures, delivering detailed observations that guided the development team in building more resilient and scalable software

Dec'17 – Jun'19: Orchestra Technology Inc. | Bellevue, WA | Client: T-Mobile

Software QA Engineer & IMS Tester for UCC Platform

Directed infrastructure resiliency and failover testing across virtualized IMS Core nodes to guarantee 24/7 service availability. Leveraged SIP/Web-RTC call flow analysis and QoS benchmarking to ensure premium user experiences in complex interworking environments.

- **Infrastructure Resiliency:** Spearheaded failover and resiliency testing for IMS Core virtual nodes, simulating large-scale network outages by disabling multi-blade configurations to ensure 24/7 service availability.
- **IMS Core Expertise:** Executed comprehensive node-level validation for Mavenir IMS elements, including Web-RTC Gateways, Application Servers (mUCCAS/VMAS), and Couch Databases (CDB).
- **Architecture & Call Flow Analysis:** Analyzed complex network architectures and SIP/Web-RTC call flows to design robust test strategies for mission-critical voice, messaging, and group calling services.
- **Cross-Technology Interworking:** Led the integration and troubleshooting between VoLTE mobile devices and Web-RTC platforms, collaborating with IMS teams to resolve high-priority interoperability bugs.
- **QoS Benchmarking:** Conducted advanced Voice and Video quality assessments using Spirent Nomad, calculating Mean Opinion Scores (MOS) to ensure premium user experience standards.
- **Product Hardening & Design Collaboration:** Partnered with Design and Network teams to implement supplementary services (Voicemail Transcript, Call Forwarding), while driving "Network Hardening" initiatives to improve overall system security and performance.

Sep'17 – Dec'17: Tech Mahindra Americas Inc. | Redmond, WA | Client - AT&T

Senior Test Engineer

Led revenue assurance and financial systems validation for AT&T Prepaid VoLTE, auditing cash flow integrity across PCRF and OCS nodes. Mitigated financial risk by validating critical firmware upgrades to ensure "zero-leakage" production deployments.

- **Financial Systems & Billing Integration:** Steered the validation of Accounting and Billing systems for AT&T's Prepaid VoLTE customers, ensuring high-accuracy transaction processing and revenue integrity.
- **End-to-End Cash Flow Validation:** Developed a deep functional understanding of telecom cash flow cycles, managing the complex interworking between **PCRF** (Policy), **DRA** (Routing), and **OCS** (Charging) nodes.

Passions and Personal Interests

- Endurance sports (cycling, running)
- Continuous Learning
- Early Educators
- Cinematic Experience

- **Strategic Requirements Analysis:** Led the Design Requirements Analysis (DRA) phase to translate business billing logic into robust technical test scenarios, preventing revenue leakage.
- **Production Readiness & Risk Mitigation:** Validated critical firmware upgrades and node configurations in pre-production environments to ensure "zero-downtime" deployments in live carrier networks.
- **Cross-Functional Technical Alignment:** Partnered with network architects to align charging triggers with real-time customer usage patterns, improving billing transparency and customer satisfaction.

Jan'12 – Jun'17: Nexius Insight Inc. | Redmond, WA | Client - AT&T

Network Engineer

Managed the performance strategy for large-scale subscriber migrations and led global OEM device certifications for 25+ vendors. Optimized QA workflows to achieve a 10% boost in productivity and a 25% increase in execution speed, accelerating product time-to-market.

- **Large-Scale Network Migration:** Analyzed and managed the performance impact of migrating Cricket Wireless subscribers onto the AT&T network, architecting test strategies to ensure node stability across PCRF, DRA, and OCS.
- **Production Readiness & Efficiency:** Spearheaded production-readiness testing for high-priority firmware releases, increasing overall productivity by 10% and ensuring timely delivery to global customers.
- **OEM & Vendor Management:** Collaborated with over 25 OEM vendors to certify 50+ devices and modems within the AT&T VoLTE environment, expanding the 10776 Test Plan to cover Carrier Aggregation and RRC/IRAT.
- **Throughput Optimization:** Optimized certification workflows to increase test execution rates by 25%, facilitating the accelerated market launch of multiple mobile products.
- **EPC Node Validation:** Led comprehensive regression and negative testing for core EPC nodes (MME, HSS, GW), utilizing deep-dive Call Flow Analysis to validate VoLTE services and IMS messaging.
- **Cross-Functional Product Integration:** Partnered with the DirecTV and PCRF teams to engineer and validate "Toll-Free Data" features, ensuring a seamless cross-platform user experience for millions of subscribers.